

Subgroups:

Definition: H of G is a subset of G satisfying the group axioms

Two trivial subgroups: $\{e\}$ and G

If $g \in G$, the smallest subgroup containing g is

$$\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

Ex: D_3

$$\{e\}, D_3, \{e, r, r^2 = r^{-1}\}, \{e, s_i\}$$

Normal subgroup:

A subgroup H is normal (invariant)

if: $gHg^{-1} = H \quad \forall g \in G$

$$H \triangleleft G$$

Ex: In D_3 : $\{e, r, r^{-1}\}$ is the only proper normal subgroup.

Simple and Semi-simple groups:

Simple: No nontrivial normal subgroups

Semi-Simple: No abelian normal subgroups

Coset decomposition and Lagrange's theorem

H is a subgroup of G . Cosets are a way of partitioning the group into disjoint pieces of equal size, each piece being "a shifted copy" of the subgroup.

for $\forall a \in G$

$$Ha = \{ ha \mid h \in H \}, \text{ right coset}$$

$$aH = \{ ah \mid h \in H \}, \text{ left coset}$$

Property: $Ha_i = Ha_j$ or $Ha_i \cap Ha_j = \emptyset$

Property: If H is normal

$$aHa^{-1} = H$$

$$aH = Ha$$

all left and right cosets coincide ...

Lagrange theorem

Let G be a finite group, H a subgroup

Then:

$$1) \quad G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_n; \quad a_1 = e$$

$$G = b_1H \cup b_2H \cup \dots \cup b_nH; \quad b_1 = e$$

$$2) \quad |Ha_i| = |a_iH| = |H|$$

3) The number of cosets is the index of H in G

$$|G| = n |H|$$

$$n = |G:H|$$

~~Ex~~ $D_3 = \{ e, r, r^{-1}, \sigma_1, \sigma_2, \sigma_3 \}$ $|D_3| = 6$

$$H = \{ e, \sigma_1 \}, \quad |H| = 2$$

$$\Rightarrow n = |G:H| = \frac{6}{2} = 3 //$$

Let's compute left cosets

$$eH = \{e, \sigma_1\} = H$$

$$rH = \{re, r\sigma_1\} = \{r, \sigma_3\}$$

$$r^{-1}H = \{r^{-1}, \sigma_2\}$$

$$\sigma_2 H = \{\sigma_2, \sigma_2 \sigma_1\} = r^{-1}H$$

$$\sigma_3 H = rH$$

$$\sigma_1 H = eH$$

$$D_3 = \{e, \sigma_1\} \cup \{r, \sigma_3\} \cup \{r^{-1}, \sigma_2\}$$

$\Rightarrow$ for a right coset:

$$D_3 = \{e, \sigma_1\} \cup \{r, \sigma_2\} \cup \{r^{-1}, \sigma_3\}$$

Exercise

$$K = \{e, r, r^{-1}\},$$

$$\begin{aligned} & \hookrightarrow gkg^{-1} = K; \forall g \in D_3 \\ & \hookrightarrow gk = kg \end{aligned}$$

$$|K| = 3 \Rightarrow n = \frac{6}{3} = 2$$

$$eK = K$$

$$rK = \{re, rr, rr^{-1}\} = \{r, r^2, e\} = K$$

$$r^2K = \{r^2e, r^2r, r^2r^{-1}\} = \{e, r, r^2\} = K$$

$$\sigma_1 K = \{\sigma_1 e, \sigma_1 r, \sigma_1 r^{-1}\} = \{\sigma_1, \sigma_3, \sigma_2\}$$

$\vdots$
 $\vdots$
 $\vdots$

$$\sigma_2 K = \sigma_3 K = \sigma_1 K = \{\sigma_1, \sigma_2, \sigma_3\}$$

$$D_3 = K \cup \sigma_1 K$$

rotations

reflections

Conjugate class decomposition

Conjugate elements: $a, b \in G$, b is conjugate of a if there exists $g \in G$
$$b = g a g^{-1}$$

Prop:

① $a = e a e^{-1}$

② If $b = g a g^{-1}$
 $\hookrightarrow a = g^{-1} b g$

③ If $b = g_1 a g_1^{-1}$; $c = g_2 b g_2^{-1}$
$$c = g_2 g_1 a (g_2 g_1)^{-1}$$

Conjugacy classes:

The conjugacy class of an element $a \in G$ is the set

$$(a) = \{ g a g^{-1} \mid g \in G \}$$

Prop:

1) If $(a) = \{ a \}$, then a is self conjugate

2) In any abelian groups, every element is self-conjugate $[g a g^{-1} = a]$

3) Conjugacy classes are invariant under conjugation: $g(a)g^{-1} = (a) \quad \forall g \in G$

Ex: D_3

$$(e) = \{ e \}$$

← exercise

$$(r) = (r^{-1}) = \{ r, r^{-1} \}$$

$$(\sigma_1) = (\sigma_2) = (\sigma_3) = \{ \sigma_1, \sigma_2, \sigma_3 \}$$

Theorem: Every group G is the disjoint union of the conjugacy classes of its elements

$$G = \bigcup_{a \in G} (a)$$

$$D_3 = \{ e \} \cup \{ r, r^{-1} \} \cup \{ \sigma_1, \sigma_2, \sigma_3 \}$$

Quotient group: H is a normal subgroup of G ($H \triangleleft G$)

Assume $G = a_1 H \cup a_2 H \cup \dots \cup a_n H$, $a_1 = e$

The set $G/H = \{a_1 H, a_2 H, \dots, a_n H\}$ is called the quotient group of G by H

Prop:

1) For $a_i H, a_j H \in G/H$

The composition is defined as $(a_i H)(a_j H) = (a_i a_j) H$

2) $eH = He = H$

3) $(a_i H)^{-1} = a_i^{-1} H$

4) $|G| = n_G$, $|H| = n_H \Rightarrow |G/H| = \frac{n_G}{n_H}$

Ex: D_3 $H = \{e, r, r^{-1}\}$

$$eH = rH = r^{-1}H$$

$$\sigma_1 H = \sigma_2 H = \sigma_3 H = \{\sigma_1, \sigma_2, \sigma_3\}$$

$$D_3/H = \{H, \sigma_1 H\}$$

D_3/H	H	$\sigma_1 H$
H	$(H)(H) = H$	$(H)(\sigma_1 H) = \sigma_1 H$
$\sigma_1 H$	$(\sigma_1 H)(H) = \sigma_1 H$	$(\sigma_1 H)(\sigma_1 H) = \sigma_1 \sigma_1 H = H$

D_3/H	H	$\sigma_1 H$
H	H	$\sigma_1 H$
$\sigma_1 H$	$\sigma_1 H$	H

$$C_2 = \{e, g\}, g^2 = e$$

$$D_3/H \simeq C_2$$

Centraliser °

the centraliser of S in G is the set of $C_G(S) = \left\{ g \in G \mid gs = sg \text{ for all } s \in S \right\}$

Normalizer °

$$N_G(S) = \left\{ g \in G \mid gSg^{-1} = S \right\} = \left\{ g \in G \mid gsg^{-1} \in S \text{ for all } s \in S \right\}$$

Center °

Center of G : $Z(G) = \left\{ z \in G \mid zg = gz \text{ for all } g \in G \right\}$

Remarks

- 1) $C_G(S)$ and $N_G(S)$ are subgroups of G
- 2) $Z(G)$ is always abelian subgroup of G
- 3) If S is itself a subgroup then $C_G(S)$ measures how close S is to being central - whereas $N_G(S)$; how close S is to being normal.

Direct Product

A group G is the direct product of H and K , $G = H \times K$ if °

- 1) Every $h \in H$ commutes with $k \in K$, $hk = kh$
- 2) Every $g \in G$ can be written uniquely as $g = hk$

Prop.

$$|H| = n_H, |K| = n_K$$

$$\hookrightarrow |G| = n_H \times n_K$$

Composition

$$(h_i, k_j)(h_e, k_m) = (h_i h_e, k_j k_m)$$

Ex ° $G = D_3 \times \mathbb{Z}_2^+$ $\mathbb{Z}_2^+ = \{0, 1\}$, addition

$\hookrightarrow$ exercise ° show $D_6 = D_3 \times \mathbb{Z}_2^+$